

Multiple Choice

1. **Answer:** C

Options: A) Fatty acids can be made from and converted to carbohydrates and amino acids; B) Carbohydrates can be made from and converted to amino acids and fatty acids; C) Fatty acids can be made from but not converted to carbohydrates and amino acids; D) Amino acids can be made from and converted to fatty acids

Note: The correct answer highlights the unique metabolic pathways of fatty acids, which can be synthesized from carbohydrates and amino acids through specific processes like gluconeogenesis, but cannot be converted back to carbohydrates, emphasizing the one-way direction of energy flow and the integration of different macronutrients in metabolism.

2. **Answer:** A

Options: A) PUFA; B) MUFA; C) SFA; D) Protein

Note: The correct answer is PUFA because both northern and southern European countries tend to have similar dietary patterns that include sources of polyunsaturated fatty acids, such as fish and vegetable oils, leading to comparable intake levels of this nutrient.

3. **Answer:** C

Options: A) High density lipoproteins; B) Very low density lipoproteins; C) Chylomicrons; D) Low density lipoproteins

Note: Chylomicrons are the lipoproteins produced in the intestines to transport dietary fats from the exogenous pathway into the bloodstream.

4. **Answer:** C

Options: A) 75-95%; B) 50- 75%; C) > 95%; D) < 50%

Note: Fats are predominantly composed of triacylglycerols, which account for more than 95% of their total composition, making this the correct answer.

5. **Answer:** C

Options: A) Apple; B) Slice of bread; C) Cheese burger; D) Broccoli

Note: A cheeseburger typically offers a high number of calories along with essential nutrients such as protein, fats, and vitamins, making it a cost-effective choice in terms of price per calorie compared to other foods.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Research has shown that elevated levels of insulin-like growth factor-1 (IGF-1)

can promote cell proliferation and inhibit apoptosis, potentially leading to the development and progression of prostate cancer.

7. **Answer:** False

Options: A) True; B) False

Note: Endurance type exercise training actually increases the liver's ability to store glycogen, thereby maintaining higher glycogen levels during prolonged activity.

8. **Answer:** True

Options: A) True; B) False

Note: Urea is the primary nitrogenous waste product of protein metabolism in humans, making it the main nitrogenous compound found in urine.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because global health statistics indicate that approximately 13% of the world's adult population is classified as obese, reflecting a significant public health concern.

10. **Answer:** True

Options: A) True; B) False

Note: Weight loss greater than 10% or failure to regain birth weight by day 14 indicates potential feeding issues or underlying health problems in newborns, which could jeopardize their growth and development.

Long Form Response

11. **Answer:** Screening for malnutrition upon hospital admission is crucial because it helps identify patients who may be at risk of poor health outcomes due to inadequate nutrition. The MUST tool, which stands for Malnutrition Universal Screening Tool, is an effective method used in this process. It assesses a patient's body mass index, weight loss, and the effects of acute illness on dietary intake, allowing healthcare providers to determine the level of nutritional risk. By using the MUST tool, hospitals can implement early interventions, improving recovery rates and overall patient care. This proactive approach ensures that patients receive the necessary nutritional support during their stay, promoting better health outcomes.

Note: correct because it emphasizes the critical role of malnutrition screening in enhancing patient care and outlines how the MUST tool effectively evaluates nutritional risk factors to improve health outcomes upon hospital admission.

12. **Answer:** Vitamin B₁₂ plays a crucial role in the body by aiding in the synthesis of methionine from homocysteine, an important process for maintaining overall health. This conversion is essential because methionine is an amino acid that contributes to protein synthesis and is a precursor for other vital compounds, including S-adenosylmethionine, which is involved in methylation reactions in the body. When homocysteine levels rise, it can lead to various health issues, including cardiovascular disease. Therefore,

adequate levels of vitamin B₁₂ are important to ensure that this conversion occurs efficiently, helping to regulate homocysteine levels and promote healthy bodily functions. *Note:* correct because it accurately describes the critical role of vitamin B₁₂ in converting homocysteine to methionine, emphasizing its importance for protein synthesis and the prevention of health issues like cardiovascular disease due to elevated homocysteine levels.

End of Answer Key